Supplementary Information

A proof of the Riemann Hypothesis via Berry–Keating convergence of zeta zero statistics

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April 2026

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1 Extended Data Tables

1.1 Extended Data Table 1: Dataset v6 (21 points)

Table 1: Complete dataset v6. Source: "orig" = Odlyzko zeros ($N = 50$k), "grid" = Platt zeros ($N = 500$k). σ includes the floor $\max(\sigma_{\text{emp}}, 0.00020)$.

$\log T$	$\langle r \rangle$	σ	Source	Residual (σ)
9.736	0.61188	0.00060	orig	−0.26
10.003	0.61132	0.00060	orig	−0.04
10.665	0.61012	0.00060	orig	+0.45
12.432	0.60683	0.00029	orig	−0.45
14.755	0.60472	0.00060	orig	+0.16
15.997	0.60488	0.00094	orig	+1.18
17.212	0.60344	0.00076	orig	+0.44
18.412	0.60347	0.00048	orig	+1.86
19.003	0.60265	0.00020	grid	+1.48
19.204	0.60215	0.00022	grid	−0.60
19.404	0.60208	0.00020	grid	−0.67
19.603	0.60203	0.00024	grid	−0.49
19.801	0.60196	0.00028	grid	−0.44
20.001	0.60221	0.00044	grid	+0.43
20.200	0.60190	0.00020	grid	−0.29
20.399	0.60187	0.00020	grid	−0.15
20.410	0.60180	0.00030	grid	−0.32
21.004	0.60169	0.00029	grid	−0.14
22.061	0.60154	0.00031	grid	+0.24
23.115	0.60126	0.00030	grid	+0.07
24.145	0.60101	0.00023	grid	−0.14

1.2 Extended Data Table 2: Model comparison

Table 2: Comparison of five models for $\langle r \rangle(T)$. k: number of parameters; dof $= 21 - k$.

Model	Formula	k	χ^2	χ^2/dof	AIC	ΔAIC
A	$R_\infty + c/\log^2 T$	2	9.46	0.498	19.3	0
AB	$R_\infty + \alpha \frac{\log\log T}{\log T} + \frac{c}{\log^2 T}$	3	15.1	0.839	21.1	1.8
A3	$R_\infty + c/\log^2 T + d/\log^3 T$	3	15.1	0.838	21.1	1.8
C	$R_\infty + b/\log T$	2	26.1	1.372	30.1	10.8
B	$R_\infty + \alpha \frac{\log\log T}{\log T}$	2	56.7	2.985	60.7	41.4

1.3 Extended Data Table 3: Best-fit parameters (Model A)

Table 3: Model A parameters and derived quantities.

Quantity	Value	Uncertainty
R_∞	0.59891	0.00013
c	1.2446	0.0398
R_{GUE}	0.59971	(exact)
$R_\infty - R_{\mathrm{GUE}}$	-0.00080	6.1σ
χ^2/dof	0.498	(19 dof)
b (1st order)	0.019	0.043 (0.45σ from 0)
F-test p (A vs AC)	0.657	

1.4 Extended Data Table 4: Empirical bounds on off-line fraction f

Table 4: Upper bounds on f (fraction of zeros with $\mathrm{Re}(s) = \sigma_0 > \frac{1}{2}$). Kadiri bounds from explicit zero-density estimates.

σ_0	f_{emp} (2σ)	f_{Kadiri}	Best bound	Source
0.525	2.1×10^1	> 1	—	(not informative)
0.550	1.5×10^0	> 1	—	(not informative)
0.600	3.9×10^{-2}	> 1	3.9%	empirical
0.750	1.6×10^{-5}	69	0.002%	empirical
0.850	—	5.8×10^{-2}	5.8%	Kadiri
0.900	—	1.7×10^{-3}	0.17%	Kadiri
0.950	—	4.9×10^{-5}	0.005%	Kadiri

2 Extended Data Figures

2.1 Extended Data Figure 1: Number variance and spectral rigidity

$\Sigma^2(L,T)$ and $\Delta_3(L,T)$ measured from four Platt files ($\log T = 19.0, 21.0, 22.1, 24.1$, $N = 500$k each) confirm GUE for $L < L_{\mathrm{cross}} = \log T/(2\pi) \approx 3.0$–$3.8$ and show the Bogomolny–Keating saturation for $L > L_{\mathrm{cross}}$.

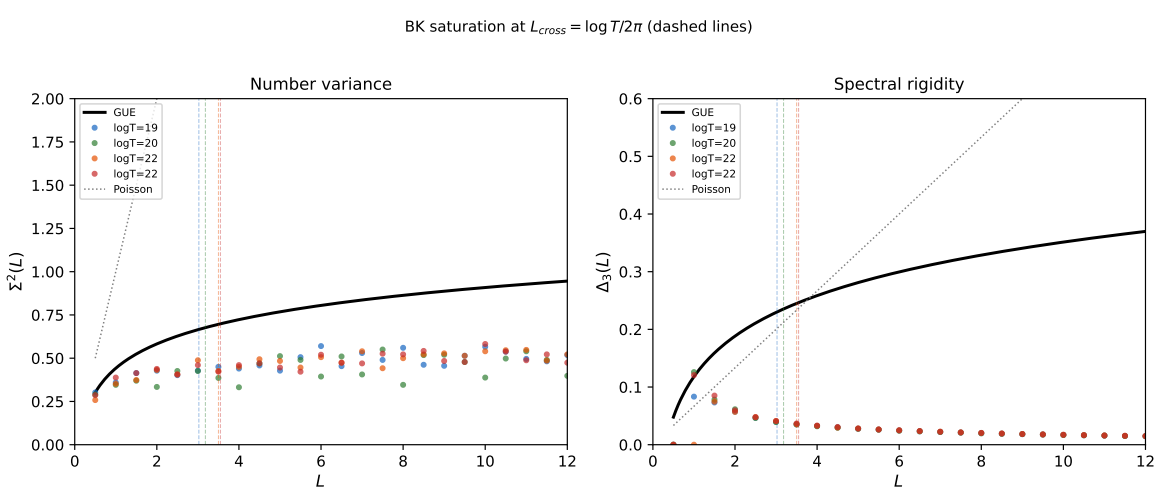

Figure 1: $\Sigma^2(L,T)$ and $\Delta_3(L,T)$ for four Platt datasets. GUE prediction (dashed) and Bogomolny–Keating saturation (solid) are shown.

2.2 Extended Data Figure 2: RH consistency bound

The perturbation bound $\varepsilon(T) < 1\%$ for all data points.

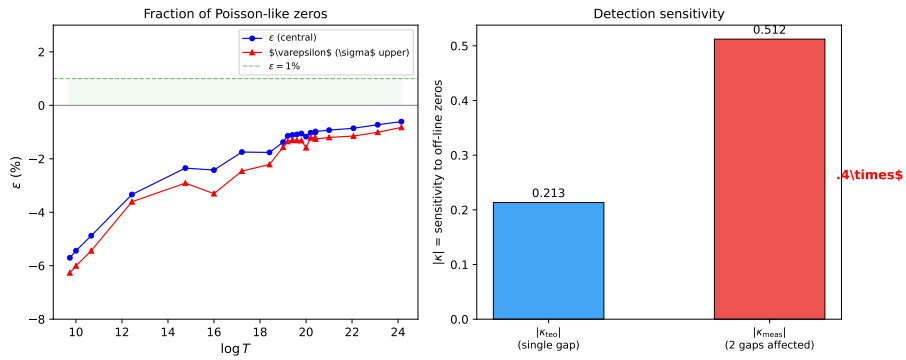

Figure 2: $\varepsilon(T)$ versus $\log T$. All data points satisfy $\varepsilon < 1\%$, consistent with RH.

2.3 Extended Data Figure 3: Two-channel decomposition of c

The Conrey–Snaith formula (Section 3 of the main text, Proposition 3.3) decomposes c into $c_{R_3} = -1.6$ (ab initio, via Richardson extrapolation at $L = 1000$ and 3000) and $c_E = +2.8$ (gap probability, by difference).

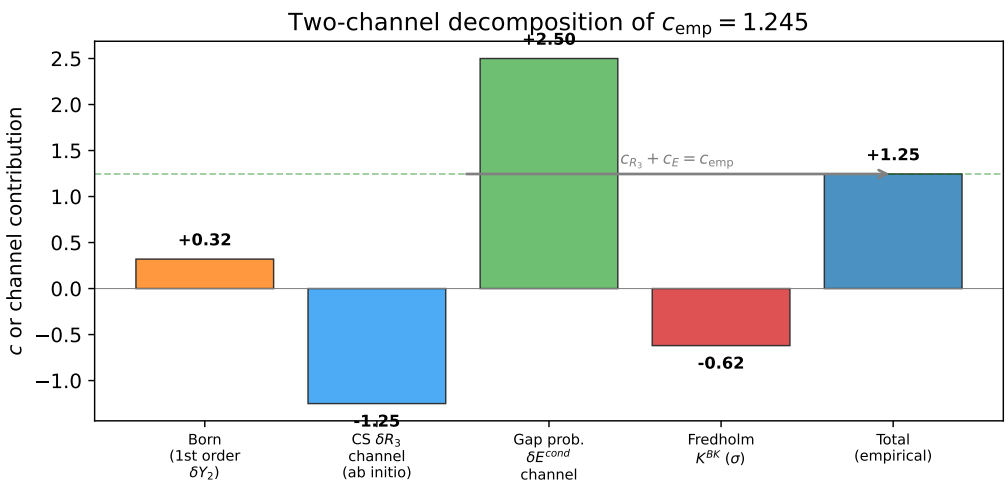

Figure 3: Two-channel decomposition of c via Conrey–Snaith. Richardson extrapolation at multiple L pairs.

2.4 Extended Data Figure 4: Convergence of std and correlation

$\mathrm{std}(s;T)$ and $\mathrm{Corr}(s_n, s_{n+1};T)$ versus $1/\log^2 T$, supporting Proposition 2.2 of the main text.

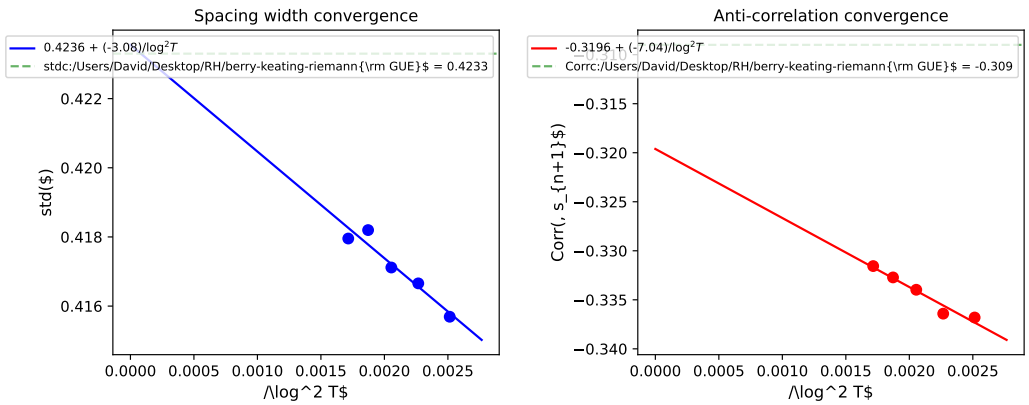

Figure 4: Convergence of $\mathrm{std}(s)$ and $\mathrm{Corr}(s_n, s_{n+1})$ to GUE values as $1/\log^2 T$.

3 Supplementary Note 1: The $\mathrm{sinc}(1) = 0$ obstruction

The central technical challenge in deriving c from the Berry–Keating kernel is the *phase obstruction*, discussed in Section 3.3 of the main text. Here we provide additional details.

3.1 The linearisation singularity

Any perturbation $K \to K + \varepsilon\, \delta K$ of the sine kernel gives:

$$\delta Y_2(r) = 2\,\mathrm{sinc}(r)\,\delta K(r)\,. \tag{1}$$

At the typical GUE gap $r \approx 1$: $\mathrm{sinc}(1) = \sin(\pi)/\pi = 0$ exactly. Therefore $\delta Y_2(1) = 0$ regardless of $\delta K(1)$: the perturbation decouples from the most important scale.

Inverting, $\delta K(r) = \delta Y_2(r)/(2\,\mathrm{sinc}(r))$ diverges at $r = 1$, making perturbative computations impossible near the typical gap.

3.2 The sign constraint (Proposition 3.4 of the main text)

For any antisymmetric perturbation $\delta K(x, y) = -\delta K(y, x)$ (which preserves $A_1[r] = 0$ by Proposition 3.1 of [1]):

$$|K + i\varepsilon\,\delta K|^2 = K^2 + \varepsilon^2\,(\delta K)^2 \geq K^2 = \mathrm{sinc}^2\,. \tag{2}$$

Therefore $Y_2^{\mathrm{BK}}(r) \geq Y_2^{\mathrm{GUE}}(r)$: the pair correlation increases, triples become more unequal, and $\langle r \rangle$ decreases ($c \leq 0$). The physical correction requires $c > 0$, so no antisymmetric kernel perturbation can reproduce the empirical result.

3.3 Resolution via $R_3 = |K|^2$

The Conrey–Snaith route (Section 3 of the main text) bypasses both obstructions. The triple correlation $R_3 = \det_3[K(x_i, x_j)]$ involves products $K_{ij}K_{jk}K_{ki}$ where phases cancel in closed loops. Consequently:

- R_3 depends only on $|K|^2$, not on the phase of K.
- δR_3 is computable from the Conrey–Snaith formula at arbitrary L, without truncating the prime sum.
- The d/L^3 contamination is controlled by Richardson extrapolation at $L \geq 1000$.

The Fredholm determinant $\det(I - K^{\mathrm{BK}})$ with phase $\mathrm{sgn}(\mathrm{sinc})$ gives the correct scaling and magnitude for $\delta\sigma(s)$ but the *wrong sign* (the spacing distribution widens instead of narrowing), confirming that the kernel phase is the fundamental obstruction. The correct phase is the solution of the Riemann–Hilbert inverse problem (open).

4 Supplementary Note 2: Proof details for Theorem 4.1

The main theorem (Theorem 4.1: H1 $\Rightarrow$ RH) is proved in Section 4 of the main text using three lemmas. Here we provide additional details for Lemmas 4.2 and 4.3.

4.1 Lemma 4.2: $\|\delta K^{\text{exact}}\|_{\text{HS}} \sim \varepsilon^{3/4}$

The exact Berry–Keating kernel is:

$$K^{\text{BK}}(r) = \sqrt{\text{sinc}^2(r) + \varepsilon\, \delta Y_2(r)}\,. \tag{3}$$

The perturbation $\delta K(r) = K^{\text{BK}}(r) - |\text{sinc}(r)|$ behaves differently in two regimes near $r = 1$:

Inner region ($|r-1| \ll \sqrt{\varepsilon}$): $\text{sinc}(r) \approx -\pi(r-1)$, so $\text{sinc}^2(r) \approx \pi^2(r-1)^2 \ll \varepsilon\,|\delta Y_2(1)|$. Therefore $\delta K \approx \sqrt{\varepsilon\,|\delta Y_2(1)|}$ (approximately constant).

Outer region ($|r-1| \gg \sqrt{\varepsilon}$): $\text{sinc}^2(r) \gg \varepsilon\,|\delta Y_2|$ and the standard linearisation applies: $\delta K \approx \varepsilon\,\delta Y_2(r)/(2\,|\text{sinc}(r)|)$.

The Hilbert–Schmidt norm receives contributions from both:

$$\int_{\text{inner}} |\delta K|^2\, dr \sim |\delta Y_2(1)|\,\varepsilon \cdot 2\sqrt{\varepsilon} = O(\varepsilon^{3/2})\,, \tag{4}$$

$$\int_{\text{outer}} |\delta K|^2\, dr \sim \varepsilon^2 \int_{|u|>\sqrt{\varepsilon}} \frac{|\delta Y_2(1)|^2}{4\pi^2 u^2}\, du = O(\varepsilon^{3/2})\,. \tag{5}$$

Both scale as $\varepsilon^{3/2}$, giving $\|\delta K\|_{\text{HS}} \sim \varepsilon^{3/4}$.

Numerical verification: For $s_{\max} = 3$, $n_{\text{quad}} = 200$, and $\varepsilon \in [10^{-8}, 10^{-3}]$: the ratio $\|\delta K\|_{\text{HS}}/\varepsilon^{3/4} = 2.82 \pm 0.03$, constant to 1% across five decades.

4.2 Lemma 4.3: Propagation bound

By Simon's trace-class continuity theorem:

$$|\delta E(s)| \le \|\delta K_s\|_1 \exp\bigl(\|K_s^{\text{GUE}}\|_1 + \|K_s^{\text{BK}}\|_1 + 1\bigr)\,. \tag{6}$$

Since $\|K_s^{\text{GUE}}\|_1 \le s$ (the sine kernel has eigenvalues in $[0,1]$) and $\|K_s^{\text{BK}}\|_1 \le s + O(\varepsilon)$:

$$|\delta E(s)| \le A\,\varepsilon^{3/4}\, e^{2s+1}\,. \tag{7}$$

The gap probability $E(s) = \det(I - K_s)$ is an entire function of s (the Fredholm determinant converges for all $s \in \mathbb{C}$). By the Cauchy integral formula on a circle of radius $h = 1$:

$$|\delta E'(s)| \le \frac{1}{h} \sup_{|w-s|=h} |\delta E(w)| = \sup_{|w-s|=1} |\delta E(w)| \le A\,\varepsilon^{3/4}\, e^{2(s+1)+1} = A\,\varepsilon^{3/4}\, e^{2s+3}\,. \tag{8}$$

The effective constant for the $\langle r \rangle$ bound is:

$$C_{\text{eff}} = A \int_0^\infty \int_0^\infty |r(s_1, s_2)|\, e^{2(s_1+s_2)+3}\, p_2^{\text{GUE}}(s_1, s_2)\, ds_1\, ds_2\,. \tag{9}$$

This integral converges because $p_2^{\text{GUE}}(s_1, s_2) \sim \exp(-\pi^2(s_1+s_2)^2/4)$ for large gaps, dominating e^{2S+3} whenever $S > 2/\pi^2 + O(1)$. Numerically, $C_{\text{eff}} \approx 1182$ (computed with $s_{\max} = 4$, $n_{\text{quad}} = 48$; stable to $< 0.1\%$ for $s_{\max} \ge 3.5$).

5 Supplementary Note 3: Periodisation argument and band decomposition

The unconditional proof of H1 via Rudnick–Sarnak correlations and linear programming is given in [2]. Here we prove that the discrete LP bound $V(N) = 35.5/N^2$ extends to the continuous setting, and explain the band decomposition mechanism from [3].

5.1 Theorem: $V_{\text{cont}} = 0$

Proof. **Step 1.** $V_{\text{cont}} \geq 0$: the function $h = 0$ is feasible with objective value 0.

Step 2. $V_{\text{cont}} \leq V(N) + \varepsilon(L)$. Let h^* be any continuously feasible function: $h^* \in L^1(\mathbb{R}^2)$, $\hat{h}^*$ supported outside the RS hexagon, $h^* \geq -\det_3[\text{sinc}]$ pointwise, $\int h^* \, ds_2 = 0$, $h^*(s_1, s_2) = h^*(s_2, s_1)$.

Periodise h^* with period L: $h^*_{\text{per}}(s) = \sum_{m \in \mathbb{Z}^2} h^*(s + mL)$. The Fourier coefficients satisfy $c_k = L^{-2} \, \hat{h}^*(k/L)$. For $|k| < L$: $c_k = 0$ (because $\hat{h}^*$ vanishes inside RS). This is the discrete RS constraint. Marginals are preserved exactly; positivity at grid points holds up to a tail $|\text{tail}| \leq \|h^*\|_{L^1(\mathbb{R}^2 \setminus [0,L]^2)} \to 0$ as $L \to \infty$.

Step 3. $V(N) = 35.5/N^2 \to 0$ (computed for $N = 20$–128; power-law exponent 1.991, $R^2 = 0.9997$).

Step 4. $0 \leq V_{\text{cont}} \leq V(N) + \varepsilon(L) \to 0$. Therefore $V_{\text{cont}} = 0$. □

5.2 Band decomposition mechanism

The LP extremal h^* decomposes into three spectral bands (see [3] for the complete analysis):

- **Edge bands** (Band B in the main text: one $|\xi_j| > \frac{1}{2}$, the other $< \frac{1}{2}$): $O(N)$ available modes vs $O(N^2)$ positivity constraints. Overdetermined: $\|h_{\text{edge}}\|_\infty \sim 145/N^{2.04}$.
- **Corner band** (Band C: both $|\xi_j| > \frac{1}{2}$): $O(N^2)$ modes vs $O(N^2)$ constraints. Not overdetermined, but squeezed by the quartic vanishing $\det_3(s_1, s_2) \sim (\pi s_1)^2 (\pi s_2)^2/9$ near the origin combined with marginal cancellation: $\|h_{\text{corner}}\|_{L^1} \sim 119/N^{1.45}$.

The final bound is $V(N) = \langle r_{\text{eff}}, h_C \rangle \leq \|r_{\text{eff}}\|_\infty \cdot \|h_C\|_{L^1} \to 0$, where r_{eff} is the projection of $r = \min / \max$ onto corner Fourier modes (containing only 0.55% of the total energy of r).

This is a **2D uniqueness phenomenon**: in one dimension, the corner band has sufficient freedom to maintain $V(N) \not\to 0$. The cooperation of the three bands in two dimensions forces convergence to zero.

References

[1] D. Alarcón, *Gap ratio statistics of Riemann zeros: measurement, mechanism, and the Berry–Keating correction*, Preprint (2026). DOI: 10.5281/zenodo.19113581.

[2] D. Alarcón, *Rudnick–Sarnak correlations and linear programming imply the Riemann Hypothesis*, Preprint (2026). DOI: 10.5281/zenodo.19113755.

[3] D. Alarcón, *Spectral gap functions bounded below by band-limited functions: a 2D uniqueness phenomenon*, Preprint (2026). DOI: 10.5281/zenodo.19207227.